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QUESTION PAPER

Name of the Examination: WINTER 2022-2023–CAT-1

Course Code: CSE-2008
Set number: 4
Duration: 90Mins

Course Title: Operating System
Date of Exam: 20-02-2023 (FN)
Total Marks: 50 (G1)

Instructions:

1. Answer All the questions

Q1. With a neat sketch, describe the services that an operating system provides to users, processes and other systems. **[10M]**

Q2. Explain different categories of System calls with suitable examples. **[10M]**

Q3. What are the advantages of inter-process communication? How communication takes place in a shared-memory environment? Explain. **[10M]**

Q4. Assume the following workload in a system: **[10M]**

Process	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

Draw a Gantt chart illustrating the execution of these jobs using the round-robin Scheduling algorithm, as well as the average waiting time and turnaround time and add the columns in table CT, TAT, and WT. Assume that the time quantum is 2 seconds.

Q5. Consider the set of 5 processes whose arrival time and burst time are given below **[10M]**

Process	Arrival Time	Burst Time
P1	3	4
P2	5	3
P3	0	2
P4	5	1
P5	4	3

If the CPU scheduling policy is FCFS and non-preemptive SJF, calculate the average waiting time and average turnaround time.

QP MAPPING

Q. No.	Module Number	COMapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1,2	1	2	1	10
Q2	1	1	1,2	2	1	10
Q3	2	2	1,2	2	1	10
Q4	2	2	4	-	1	10
Q5	2	2	4	-	1	10

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QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-1

Course Code: CSE2008
Course Title: OPERATING SYSTEMS
Set number: 5
Date of Exam: 20-02-2023 (AN) (G2)
Duration: 90 Min.
Total Marks: 50 M

Instructions:

1. Answer All the questions

Q1. a. Discuss in detail about Distributed Operating system, Real Time Embedded Systems, Open-source Operating Systems. [5M]

b. What is the purpose of the system calls and explain. [5M]

Q2. a. Explain the process of fork systems call with diagram. [5M]

b. What are the major activities of an operating system in regard to process management and memory management? [5M]

Q3. What do you mean by PCB? Where is it used? What are its contents? Explain. [10M]

Q4. Consider the following data to evaluate average waiting time and turnaround time using Non-preemptive SJF and Round Robin CPU scheduling algorithms. Assume time slice is 3. [10M]

Process	Burst Time	Arrival Time
P1	6	0
P2	7	2
P3	4	4
P4	10	5
P5	5	6

Q5. Consider the following data to evaluate average waiting time and turnaround time using FCFS & SJF (preemptive) algorithm and mention the disadvantages of FCFS algorithm. [10M]

Process	Arrival Time	Burst Time
P1	0	10
P2	1	5
P3	2	3
P4	3	2

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2	1	1	10
Q2	1	1	1,2,4	2,3	1,2	10
Q3	2	2	1	2	1	10
Q4	2	2,5	2,5	4	1,2,4,5	10
Q5	2	2,5	2,5	4	1,2,4,5	10

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QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-1

Course Code: CSE2008

Course Title: Operating Systems

Set number: 7

Date of Exam: 16-02-2023 (FN) (D1)

Duration: 90 Min

Total Marks: 50

Instructions:

- *No clarification during the examination; answer the questions as per your best understanding (carry marks).*
- *Answer all questions.*

Q1. a. What is the role of the bootstrap program? Where does it been stored? **[5M]**

b. How is the distributed operating system different from the network operating system? **[5M]**

Q2. a. Name any three system calls responsible for process control activities. Write the proper syntax for each of them for Windows and Unix operating systems. **[5M]**

b. Write a simple C program that illustrates the use of these three system calls. **[5M]**

Q3. a. Other than the program itself, what are the different additional things required by a program to become a process? Also, elaborate on the contents of a PCB. **[5M]**

b. Explain shared memory concept of IPC considering the 'producer–consumer' problem. **[5M]**

Refer to the following table to answer Q4 and Q5.

Process	Arrival Time	CPU Burst Time
P0	0	15
P1	1	15
P2	1	25
P3	2	5

Q4. Draw Gantt charts and compute the average waiting time and turn-around time following preemptive and non-preemptive SJF CPU scheduling algorithms. **[10M]**

Q5. Apply the Round-Robin CPU scheduling algorithm (Time quantum is 2 ms) and comment on improvement in performance over the preemptive SJF (that you got after solving the Q4) concerning the average waiting time and turn-around time. **[10M]**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1	1	10
Q2	1	2	1, 2, 4	2, 3	1, 2	10
Q3	2	3	1, 2, 3, 5	1, 3	1, 2, 4	10
Q4	2	4	2, 5	4	1, 2, 3, 4	10
Q5	2	4	2, 5	4	1, 2, 3, 4	10

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QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-1

Course Code: CSE2008 | **Course Title: Operating Systems**
Set number: 9 | **Date of Exam:** 16-02-2023 (AN) (D2)
Duration: 90 Min | **Total Marks: 50M**

Instructions:

1. Answer all the questions

Q1. a. "Operating System is a Resource Manager and Control Program". Explain this statement with suitable functionality of Operating System. **(5M)**

b. List the advantages of peer-to-peer systems over client-server systems? **(5M)**

Q2. Explain different categories of system calls with suitable example. **(10M)**

Q3. Consider the set of processes with arrival time and the length of the CPU burst time given in milli seconds.

Process ID	Arrival Time	Burst Time
P1	0	3
P2	1	1
P3	2	3
P4	3	6
P5	4	2

Draw the Gantt chart, calculate and compare the average waiting time and average turnaround time for the following CPU Scheduling algorithms.

a) Shortest Job First (Non-Preemptive)
b) Round Robin (Time Quantum – 3ms) **(10M)**
c)

Q4. Analyze the execution of the following processes using Priority scheduling algorithm preemptive and non preemptive mode through Gantt chart and calculate the average waiting time and average turnaround time.

	Burst Time	Priority	Arrival Time
P1	4	2	0
P2	3	1 (High)	0
P3	7	4	3
P4	5	3	4
P5	6	5 (Low)	5

(10M)

Q5. Explain direct and indirect communications of message passing system with suitable examples. **(10M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1	1	1	10
Q2	1	2	1	1	1	10
Q3	2	4	2	1	1	10
Q4	2	4	2	1	1	10
Q5	2	3	1	1	1	10